

ANISH NAIR

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A1

Home Work

Method Overriding

A vehicle tracking system needs to calculate the maximum speed of different types of vehicles.

- A generic vehicle has a base speed.
- A car has a higher max speed.
- A bike has an even higher max speed.

Method Overloading

Scenario: Payment Processing System

A company is building a payment gateway that allows users to pay using different methods:

- Credit Card (card number & CVV)
- Bank Transfer (account number & IFSC code)
- UPI (UPI ID)

VS code Folder stuff > J Main.java

```
1  class Vehicle {
2      int getMaxSpeed() {
3          return 100;
4      }
5  }
6
7  class Car extends Vehicle {
8      int getMaxSpeed() {
9          return 180;
10     }
11 }
12
13 class Bike extends Vehicle {
14     int getMaxSpeed() {
15         return 220;
16     }
17 }
18
19 public class Main {
20     public static void main(String[] args) {
21         Vehicle v1 = new Vehicle();
22         Vehicle v2 = new Car();
23         Vehicle v3 = new Bike();
24
25         System.out.println(v1.getMaxSpeed());
26         System.out.println(v2.getMaxSpeed());
27         System.out.println(v3.getMaxSpeed());
28     }
29 }
```

100

180

220

VS code Folder stuff > J Main.java

```
1 public class Main {  
2  
3     static void pay(String cardNumber, String cvv) {  
4         System.out.println("Paid using Credit Card: " + cardNumber + ", CVV: " + cvv);  
5     }  
6  
7     static void pay(String accountNumber, String ifscCode, boolean isBankTransfer) {  
8         System.out.println("Paid using Bank Transfer: " + accountNumber + ", IFSC: " + ifscCode);  
9     }  
10  
11     static void pay(String upiId) {  
12         System.out.println("Paid using UPI: " + upiId);  
13     }  
14  
15     public static void main(String[] args) {  
16         pay("1234567812345678", "123");  
17         pay("9876543210", "IFSC0001", true);  
18         pay("user@upi");  
19     }  
20 }
```

```
Paid using Credit Card: 1234567812345678, CVV: 123  
Paid using Bank Transfer: 9876543210, IFSC: IFSC0001  
Paid using UPI: user@upi
```